

## EDUCATION

### Master's - Master of Applied Computer Science

Jan 2024 – Dec 2025

CONCORDIA UNIVERSITY, MONTREAL, QUEBEC, CANADA

- **Relevant Courses:** Distributed System Design | Natural Language Processing | Machine Learning | Artificial Intelligence | Programming and Problem Solving.

## SKILLS

**Programming:** Java | Python | SQL | HTML | CSS | Java Script.

**AI:** Natural Language Processing (NLP) | Transformers | LSTMs | LLMs.

**Data Science & Analytics:** Pandas | NumPy | Scikit-Learn | TensorFlow | PyTorch.

**Tools & Frameworks:** AWS | Git | IntelliJ | VS Code | Spring Boot | React Js.

**SoftSkills:** Communication | Time-management | Teamwork | Adaptability | Task Prioritization | Problem solving.

## PERSONAL AND ACADEMIC PROJECTS

### Distributed Health Care Management System Implementation Using WebService

- Implemented **3 servers (MTL/QUE/SHE)** communicating via **SOAP Web Services (JAX-WS)**.
- Built **nested ConcurrentHashMaps** and patient lists for thread-safe scheduling updates.
- **SKILLS:** Java | UDP Sockets | Web Services (SOAP) | UDP Sockets | Distributed Systems | [GitHub](#)

### Warzone Command-Line Strategy Game

- Developed a Java-based strategy game simulating territory conquest with single and tournament modes.
- Implemented **multiple design patterns (State, Observer, Command, Adapter, Strategy)** for scalable architecture.
- **SKILLS:** Java | Object-Oriented Design | Design Patterns | Git | Maven | Junit | [GitHub](#)

### Machine Translation with Transformers

- Developed a machine translation pipeline using Hugging Face Transformers on the **IWSLT 2017 English French dataset** on a **100,000-sample** dataset over **10 epochs**.
- Implemented an end-to-end pipeline covering **data preprocessing, training, hyperparameter tuning, and evaluation**.
- **Custom Transformer** achieved **0.8581 Precision / 0.8435 Recall / 0.8506 F1 / 0.2978 METEOR**; **T5** achieved **0.8960 Precision / 0.8987 Recall / 0.8972 F1 / 0.5160 METEOR**, highlighting performance gains from trained model.
- **SKILLS:** Python | Hugging Face Transformers | NLP | Machine Translation | Dataset Preprocessing | BERT Score | ROUGE | Model Evaluation | Text Normalization | [GitHub](#)

### Multi-Instance Iris Template-Based Intruder Detection Using Minkowski Distance

- The aim of the project is distinguished whether the user is intruder or valid user based on human Irsi.
- Achieved a high **accuracy rate of 98.75%** by utilizing the Minkowski distance technique compared to existing Hamming distance models.
- Integrated Daugman's algorithm for iris localization and segmentation, template generation via Logical OR operation, key generation using XOR operations on primary diagonals, and classification via the K-Nearest Neighbor (KNN) algorithm.
- **SKILLS:** Python | Machine-learning | [Publication](#)

## WORK HISTORY AND CERTIFICATIONS

### Python Programming Trainee (APSSDC)

Mar 2022 - Jun 2022

- Completed a structured internship focused on Python programming, data structures, and real-time project development.
- By using OOP, data processing techniques to automate deployments, and operational tasks.
- Strengthened problem-solving and debugging skills while gaining insight into professional software development workflows.

### Aws Cloud Intern (AICTE)

Oct 2021 - Dec 2021

- Deployed and managed scalable cloud infrastructure using core AWS services including EC2, S3, and Lambda.
- Gained hands-on experience with IAM, VPCs, and serverless architecture for secure and efficient operations.
- Designed and implemented a sample cloud-based web application demonstrating cost-optimized deployment.

**Certifications:** AWS Solutions Architect | AWS Cloud Practitioner | Wipro Talent Next (WIPRO) | SQL (HACKER RANK).